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Accounting for Decision Value: Calibration and Selection over Prediction in a Deployed Equity-Forecasting System

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Abstract: Machine-learning on equity markets mostly is graded on predictive accuracy or area under the ROC curve, and that's the reason many fail to beat a trivial drift baseline, once costs and look-ahead are handled honestly. We argue that using accuracy is measuring the wrong thing. A deployed system delivers, what is called as decision value: the quality of the small set of positions it actually takes into consideration, weighed against the far larger set it stays out of. We make this precise by extending the classical calibration–refinement and reliability–resolution decompositions of forecast quality to the case of abstention, cost-aware decision setting, and we prove that a tail–AUC decoupling result: a one-sided selective rule, can earn a positive edge in its high-conviction tail while global AUC sits at or below one half, so AUC and accuracy systematically misreport what such a system is worthy of. We then account for the value of a real and working, deployed system on the Indian National Stock Exchange across its whole pipeline, which is: multi-source features, a hybrid gradient-boosted/recurrentensemble, language-model news scoring, post-hoc calibration, a conviction gate, and an append-only audit ledger. The numbers are unforgiving and asbolute. On 621 resolved forecasts from the live ledger, nominal 90% price intervals covered only 70.7% of outcomes, so deployed confidence was slightly overstated and correcting it was no one's job. A three-stage recalibration pipeline was applied to address a domain mismatch in the stacking layer. This correction alone reduced the error for the intended calibration from 0.35 to 0.05 with virtually no drop in predictive ability. On its own, the base model has little direction when evaluated for the given names, which is reflected in the OOS accuracy (50.1%) and ranking AUC (0.49) of about no better than 50-50 chance. However, when a conviction gate is applied to the base model, the picture completely changes. If judgments are withheld on roughly 94% of names and the system acts only when uncertainty estimates allow, the 30-day hit rate rises to 60.6% (95% interval: 58.8–62.4%) against a 58.0% base rate, with this held for 4 of the 5 evaluation years. The realized edge, then, does not come from the ranking model itself, which contributes virtually no additional alpha, but from the selection mechanism it enables and uses. What makes this mechanism reliable is proper calibration, which is why without accurate uncertainty estimates, the gate cannot be trusted to abstain at the right times. The practical implication is rather straightforward. Rather than chasing improvements to a ranking metric, like accuracy, that the deployment pipeline will never directly optimize for, effort is better spent instrumenting the uncertainty layer and evaluating the system in terms of precision at a given coverage level. That is the quantity that actually governs live performance in the volatile stock markets.

Keywords: decision value, calibration, selective prediction, probabilistic forecasting, financial machine learning, model evaluation, abstention
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1 Introduction

Any person who studies and surveys the past 10 years of machine learning research applied to financial markets is likely to walk away with a nagging question in their mind, that is, almost all the studies and research papers report directional accuracies very comfortably in the high fifties or sixties, which is quite profitable, and ROC curves sitting well above the diagonal, and backtested Sharpe ratios that would, if taken at the mentioned face value, have made their authors independently wealthy and rich long ago [Sezer et al. 2020, Jiang 2021]. And yet the practitioners who actually manage capital and try using the said mechanisms, are not exploiting these inefficiencies away behind closed doors. The distance and real–world gap between, what the academic record claims, and what the industry actually deploys and uses, is wide enough that it must surely be something in how the field measures itself must be fundamentally off.

Hence, we assert that the measurement framework itself is the problem. Accuracy and AUC are answers to a question about ranking, plainly saying that, given any two assets, does the model tend to place the one that subsequently rose above the one that fell? This question has a clean, well-studied answer, but again, it is nearly the wrong one for a system that must decide, asset by asset and day by day, whether to take a position or do nothing. A forecaster in deployment should not assign grades to every name in the universe. It should commit to a handful and pass on the rest, and its practical worth lies in the quality of those commitments, which it was sure to give, relative to what was left on the table by abstaining. We refer to this as decision value. Decision value is very much distinct from accuracy, and it is not a simple monotone function or a mathematical variation of AUC. The unsettling yet true implication is that a system can carry genuine decision value even when its accuracy and AUC suggest it has none.

We are not so much interested in an attempt to estimate this exact same decision value as the one described above in an imaginary simulated environment, as we are to evaluate it in a live, deployed one. In this paper we focus on a live, equities-forecasting system that specializes in the most liquid large-cap stocks on the National Stock Exchange of India (NSE), uses many well-worn machine learning concepts found in studies of financial forecasting, but combines them in novel ways: a multi-source feature stack (price, trend, momentum, mean-reversion and volatility signals); a hybrid ensemble method (gradient-boosted trees plus recurrent sequence model); a language model layer (that processes news sentiment and flow); a post-hoc calibration step; a conviction gate that determines if a forecast should be acted upon; and an append-only forecast ledger that writes each forecast to the disk at the time it's generated (so that no one, including us, can ever go back to change the values, even so far as to save the recorded values themselves for another use). The platform has two user-facing modules, and it has been appending its forecasts to its local disk, since March 2026, prior to the arrival of the subsequent values.

The thing the live record told us is useful to put out there first since it orders all that follows. First, the calibrated uncertainties the models were outputting were consistently wrong in one direction; nominal 90% price ranges actually contained the realized price only 70.7% of the time for 621 completed forecasts, and none of the pipeline was sounding alarms. Later this miscalibration was repairable: expected calibration error decreased from 0.35 to 0.05 in repair, and (as discussed) did not touch predictive skill, it just moved our probability output values back into a range that the real world had data for. Second, and once the probabilities were a legitimate thing to analyze, the directional skill we were afraid might be masked by this miscalibration, was not actually present-for this universe of individual names, at short horizons, the next-period direction is essentially random outside of the general upward bias of equities, with out-of-sample hit rate at 50.1% and ranking AUC at around 0.49. Third, the primary driver of system value remains a conviction gate that aborts a signal on around 94% of names, but that captures a small but consistent edge when it doesn't: 60.6% 30-day hit rate for the signal, versus a 58.0% base rate, that was valid in 4 out of 5 years.

When viewed from a perspective within a traditional framework of accuracy metrics, these results can be interpreted as a sequence of failures and unusable outcomes. But viewed as a decision-value account, they tell a coherent story, one which we believe, extends well beyond this particular system. The realised value of the predictive model itself is almost none. What value exists originates from selection: the discipline of acting only in the high-conviction tail of the distribution. Calibration, in turn, is what allows the conviction threshold to mean what it claims to say. The broad upward drift of the market is beta, not skill. The ranking signal that standard metrics are designed to reward is, in this case, simply absent. Judged by the conventional scorecard, the system appears to offer nothing but garbage values. Judged by the decision value, which as we state, is more important in financial markets, it can actually be compensated for, and it exhibits a thin but real edge over all other algorithms, and one whose origin can be traced to a specific, identifiable component.

1.1 Contributions

The paper makes three contributions, one conceptual, one theoretical, and one empirical.

1. A decision-value accounting framework (Section 3). We define the decision value of a selective, cost-aware forecaster and decompose it into a drift (base-rate) term, a selection (refinement) term realised at a conviction threshold, and a calibration term that enables rather than adds. The decomposition is the selective-prediction counterpart of the calibration–refinement partition of [DeGroot and Fienberg 1983] and the reliability–resolution partition of [Murphy 1973], and it makes "where does the value come from" a question with a measurable answer.

2. A tail–AUC decoupling result (Section 3.4). We prove that the selection term is a property of the extreme tail of the score distribution and is largely independent of global ordering: there exist score–label distributions with AUC no greater than one half on which a one-sided conviction rule still earns a strictly positive tail edge. The practical consequence is that AUC and accuracy can report zero skill on a system that has decision value, which is the regime we then measure, and it is why precision at a fixed firing rate is the right instrument.

3. A whole-system attribution on a deployed platform (Sections 4–5). We instrument a live system and report measured numbers only: live-ledger interval coverage, a calibration repair that traces to a concrete stacking bug, a verified accuracy ceiling established five ways, and a walk-forward selective signal over eight years. We then carry out the attribution, separating the contribution of drift, calibration, selection and ranking, and showing that ranking contributes essentially nothing while selection carries the system.

We are careful about scope throughout, in keeping with a standing rule on this project that no reported number is synthetic. Where the live ledger cannot support a claim (most importantly, an isolated return attributable to the news layer, for which per-event labels are not stored), we say so and argue from structure and from the measured end-to-end behaviour instead of manufacturing a clean figure. Section 6 returns to these limits.

2 Related work

Machine learning for markets, and how it is graded. Deep and ensemble models for equity prediction are now routine. Long short-term memory networks [Fischer and Krauss 2018], gradient-boosted trees and random forests for statistical arbitrage [Krauss et al. 2017], and large cross-sectional studies of machine-learning asset pricing [Gu et al. 2020] are representative, and two systematic reviews map the territory [Sezer et al. 2020, Jiang 2021]. The dominant evaluation currency in this literature is directional accuracy or AUC, sometimes a backtested Sharpe ratio. A parallel and more sceptical methodological line has shown how fragile those currencies are: backtest overfitting inflates apparent performance [Bailey et al. 2017], the cross-section of "discovered" predictors is riddled with multiple-testing artefacts [Harvey et al. 2016], and a careful treatment of leakage, costs and capacity removes most reported edges [Lopez de Prado 2018]. Our paper sits inside this sceptical line but turns it constructive: rather than only warn that the usual scorecard misleads, we replace it with a decision-value account and show what the account reveals on a system we operate.

Calibration of probabilistic forecasts. Modern classifiers are frequently overconfident [Guo et al. 2017]. The standard post-hoc remedies are Platt scaling [Platt 1999] and isotonic regression [Zadrozny and Elkan 2002], with reliability assessed by the expected calibration error [Naeini et al. 2015] and the Brier score [Brier 1950, Niculescu-Mizil and Caruana 2005]. The foundational decompositions of forecast quality are older and, for our purposes, central: [Murphy 1973] partitions the Brier score into reliability, resolution and uncertainty, and [DeGroot and Fienberg 1983] partition forecaster quality into calibration and refinement. [Gneiting et al. 2007] crystallise the operating principle these decompositions imply (maximise sharpness subject to calibration), which is the thesis of this paper carried into the decision setting. We use these results as the scaffolding of our framework, not merely as background. A known trap when calibration meets stacked generalisation [Wolpert 1992] is that a calibrator fitted on one stage's scores and applied to another silently fails. We diagnose this failure in Section 4.3.

Selective prediction and abstention. Allowing a model to decline to answer is an old idea [Chow 1970] with a modern theory of risk–coverage trade-offs [El-Yaniv and Wiener 2010, Geifman and El-Yaniv 2017] and a distribution-free cousin in conformal prediction [Vovk et al. 2005, Angelopoulos and Bates 2021], including variants built for the non-stationary coverage failures we observe [Gibbs and Candès 2021]. This community measures risk at a given coverage; the financial-ML community, by contrast, rarely reports whether its probabilities are trustworthy or whether the model knows when to stay out. Our decision-value framework argues that the selective view is the correct one for deployed financial systems, and makes that argument quantitative on a real deployment. The closest framing in spirit treats trustworthy financial AI as an abstention problem; we differ by making the object of study the attribution of value across pipeline stages rather than a prescriptive contract, and by supplying the tail–AUC result that explains why the attribution comes out as it does.

Text and returns. That media content carries return-relevant information is well established [Tetlock 2007, Antweiler and Frank 2004], that finance-specific dictionaries matter because general ones mislabel domain terms is the lesson of [Loughran and McDonald 2011], and supervised or pre-trained language models now dominate sentiment extraction [Devlin et al. 2019, Ke et al. 2019]. Whether instruction-tuned models can read financial news and anticipate moves is under active study [Lopez-Lira and Tang 2023]. Our system uses a language model as one feature source; we are careful, in Section 5, not to claim an isolated alpha for it that our ledger cannot support.

Forecast value and decision-theoretic evaluation. There is a different tradition which asks not whether a forecast is accurate but whether it is useful, in other terms, the concrete gain a decision-maker derives by acting on it. Murphy [Murphy 1993] distinguished between the consistency, quality, and value of a forecast, arguing that the three need not move together, similar to an earlier statement of our concern that a high AUC need not become a usable signal all the time. Granger and Pesaran [Granger and Pesaran 2000] sharpened this point for economical studies, showing that a definable statistical loss function and the economic payoff of a decision, can rank forecasters in entirely different orders than normally. As stated, proper scoring rules formalise when honest probabilities are rewarded [Gneiting and Raftery 2007], and in finance the realised worth of an executable strategy is typically read off a risk-adjusted return such as the Sharpe ratio [Sharpe 1994]. Decision value belongs to this lineage. It is a value measure, conditioned on a cost-aware action rule, rather than a quality measure averaged over every forecast. What we add is a decomposition of that value into named, separately measurable parts, and a theorem that explains why quality measures fail to capture it. Our paper sits at the intersection of three literatures that rarely talk to one another: the score decompositions of forecast verification, the risk–coverage view of selective prediction, and a financial-ML literature that has learned to distrust its own evaluations. We combine them into an attribution method and apply it to a live system.

3 A decision-value accounting framework

This section sets up the bookkeeping. We define the value a selective forecaster delivers, decompose it into interpretable terms, and prove the result that makes the decomposition necessary rather than cosmetic: that the actionable term lives in the tail and is invisible to AUC.

3.1 Forecasts, decisions, and decision value

Fix a horizon of H trading days. For each name and date the system emits a calibrated directional probability $\hat{p} \in [0, 1]$ that the H-day forward return is positive, together with a point forecast and a price interval. Let $y \in \{0, 1\}$ be the realised direction. A one-sided selective rule d_τ acts, taking a long position for the horizon, when $\hat{p} \geq \tau$ and abstains otherwise. We focus on the one-sided long/abstain rule because, as Section 5.3 shows empirically, confident short calls on single names are destroyed by drift; nothing in the framework requires one-sidedness.

Two quantities summarise the rule's behaviour. Its firing rate $\phi(\tau) = \Pr[\hat{p} \geq \tau]$ is the fraction of opportunities on which it acts, and its fired precision $\pi(\tau) = \Pr[y = 1 \mid \hat{p} \geq \tau]$ is the realised up-rate among the positions it takes. Let $b = \Pr[y = 1]$ be the unconditional up-rate, the base that a name-blind "always act" policy would achieve.

Definition 1 (Selection value and decision value)**.** The selection value of the rule d_τ is the edge of its fired positions over the base,

$$\sigma(\tau) \;=\; \pi(\tau) - b. \tag{1}$$

Its decision value at firing rate ϕ is the selection value scaled by how often the rule is willing to act and net of the per-trade cost κ of acting,

$$V(\tau) \;=\; \phi(\tau)\left[\, g\,\sigma(\tau) - \kappa \,\right], \tag{2}$$

where g converts a unit of directional edge into a unit of payoff for the position sizing in use.

Equation (2) is intentionally minimal: the realised worth of a selective forecaster is set by how often it acts, how much its acted-upon calls beat the base, and what acting costs. Nothing else enters. Neither accuracy nor AUC appears in it. A model can be excellent on both and still have $V \leq 0$ when its edge does not survive cost; a model at chance on both can have $V > 0$ if it finds a thin, reliable tail. The cost κ and the sizing factor g are venue-specific, so the empirical attribution of Section 5 concentrates on the cost-free core of V (the selection value σ) and returns to the net-of-cost picture once σ has been measured (Section 5.4). The rest of the framework is about where σ comes from and why the usual metrics cannot see it.

3.2 Decomposing fired precision

The fired precision admits a decomposition that mirrors the classical partitions of forecast quality but is conditioned on the act region rather than averaged over all forecasts. Write the act region as $R_\tau = \{\hat{p} \geq \tau\}$ and let $\bar{p}(\tau) = \mathbb{E}[\hat{p} \mid R_\tau]$ be the mean predicted probability there. Adding and subtracting $\bar{p}(\tau)$ in (1),

$$\underbrace{\pi(\tau) - b}_{\text{selection value } \sigma(\tau)} = \underbrace{\big(\pi(\tau) - \bar{p}(\tau)\big)}_{\text{calibration gap on } R_\tau} + \underbrace{\big(\bar{p}(\tau) - b\big)}_{\text{conviction lift}}. \tag{3}$$

The second term, the conviction lift, is how far above the base the model is willing to claim on its acted-upon names; it is pure refinement, the analogue of resolution in [Murphy 1973]. The first term, the calibration gap, is how wrong that claim turns out to be on those same names; under perfect calibration it is zero and the realised edge equals the claimed lift. Miscalibration enters only through this term: a model that systematically over-claims has a negative calibration gap that eats into, or reverses, the lift it advertises. Calibration here enables the edge without adding to its size. It does not generate conviction lift, which is a property of how the model ranks opportunities. But when it is broken, the lift on paper is not the edge you collect, and the threshold τ stops meaning what you set it to.

The decomposition also explains why repairing calibration can leave accuracy untouched while transforming a system's usability. A monotone recalibration map (Platt or isotonic) is order-preserving, so it moves neither accuracy nor AUC nor the conviction lift $\bar{p}(\tau) - b$ evaluated at the matched quantile; a non-monotone calibrator could reorder names and would carry no such guarantee. What the monotone map moves is the calibration gap, pulling $\bar{p}(\tau)$ back onto $\pi(\tau)$ so that the advertised probability equals the delivered one. Section 5.2 shows this happening to a real model.

3.3 Relation to the Brier-score decomposition

The two terms in (3) are the act-region shadows of a classical result. Murphy [Murphy 1973] split the Brier score of a probabilistic forecaster into three pieces,

$$\mathrm{BS} = \underbrace{\mathbb{E}\big[(\bar{p}_{\mathrm{bin}} - \bar{y}_{\mathrm{bin}})^2\big]}_{\text{reliability}} - \underbrace{\mathbb{E}\big[(\bar{y}_{\mathrm{bin}} - b)^2\big]}_{\text{resolution}} + \underbrace{b(1-b)}_{\text{uncertainty}}, \tag{4}$$

where the expectations run over the forecaster's probability bins, $\bar{y}_{\mathrm{bin}}$ is the realised rate in a bin, and b is the base rate. Reliability is small when the forecaster is calibrated;

resolution is large when its bins pull outcomes apart; the uncertainty term depends only on the base rate, never on the model. A conviction gate keeps a single bin, the act region R_τ, and reads two of these three quantities in (4) through it. The conviction lift $\bar{p}(\tau) - b$ of (3) is the local face of resolution: a forecaster with no resolution cannot push its acted-on probabilities above the base, so the gate has nothing to fire on. The calibration gap $\pi(\tau) - \bar{p}(\tau)$ is the local face of reliability, and the expected calibration error of (7), defined in Section 4.3, is its size summed over bins. This is the formal reason the attribution falls out the way it does. Selection value is resolution restricted to the tail; calibration is reliability; and the base rate, the uncertainty term, is the drift that neither the model nor the gate may claim. We record the correspondence as a lemma.

Lemma 1 (Local resolution and reliability). *Fix τ and treat the act region R_τ as one bin, with mean forecast $\bar{p}(\tau)$ and realised rate $\pi(\tau)$. The selection value splits as $\sigma(\tau) = \big(\pi(\tau) - \bar{p}(\tau)\big) + \big(\bar{p}(\tau) - b\big)$, a local reliability gap plus a local resolution lift. If the underlying forecaster has zero resolution, so that its realised positive rate equals the base b on every set fixed by the forecast value, then $\pi(\tau) = b$ for every threshold τ, and therefore $\sigma(\tau) = 0$: no gate can extract an edge.*

Proof. The split is (3) with $\bar{p}(\tau) = \mathbb{E}[\hat{p} \mid R_\tau]$. Zero resolution means the realised rate is b on every event determined by the forecast value; $R_\tau = \{\hat{p} \geq \tau\}$ is such an event, so $\pi(\tau) = \Pr[y = 1 \mid R_\tau] = b$, whence $\sigma(\tau) = \pi(\tau) - b = 0$. □

Lemma 1 says a gate cannot manufacture an edge from a forecaster that does not separate outcomes. All it can do is harvest whatever resolution exists, which sits in the tail, and only if calibration is good enough that the tail is where the forecaster claims it is. That proviso is the entire job of the calibration repair in Section 5.2.

3.4 Selection lives in the tail: a decoupling from AUC

A decision-value account would just relabel the usual numbers if its central term tracked them. It does not. The selection value can be positive when AUC reports no signal at all. AUC is a global statistic, the probability that a random positive outscores a random negative, and it averages concordance over the whole score distribution. Selection value, by (1), depends only on the conditional up-rate inside a thin top slice. The two can come apart, and on real market data they do.

Proposition 1 (Tail–AUC decoupling). *For every $\varepsilon \in (0, \frac{1}{2})$ and every $\sigma_0 \in (0,\, 1-b)$ there is a joint distribution of a score S and label $Y \in \{0,1\}$ with base rate $\Pr[Y = 1] = b$ such that, for the threshold τ with firing rate $\phi(\tau) = \varepsilon$, the selection value satisfies $\sigma(\tau) \geq \sigma_0$, while the global $\mathrm{AUC}(S, Y) \leq \frac{1}{2}$.*

Proof. We construct such a distribution explicitly. Put all score mass on three cells, ordered $B_1 < B_2 < T$ by score, with ties allowed only within a cell. The top cell T has mass ε and conditional positive rate $b + \sigma_0$ (a valid probability since $\sigma_0 < 1 - b$); the two bulk cells B_1, B_2 each have mass $m = (1-\varepsilon)/2$ and positive rates $p_1 = b + a$ and $p_2 = b - c$. The threshold $\tau = \inf T$ fires on exactly T, so $\phi(\tau) = \varepsilon$ and $\pi(\tau) = b + \sigma_0$, hence $\sigma(\tau) = \sigma_0$. Holding the overall base at b requires $\varepsilon(b + \sigma_0) + m(p_1 + p_2) = b$, i.e.

$$a - c = -\frac{2\varepsilon\sigma_0}{1-\varepsilon}. \tag{5}$$

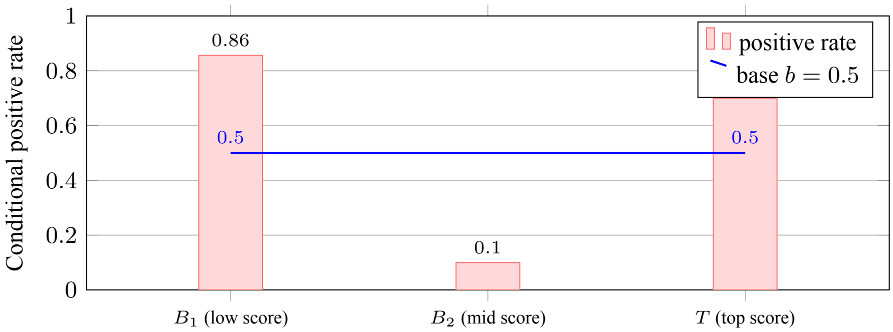

Figure 1: The tail–AUC construction of Proposition 1, drawn for the worked instance ($b = \frac{1}{2}$, $\sigma_0 = 0.2$, $\varepsilon = 0.1$, $c = 0.4$). The bulk is anti-sorted: the low-score cell B_1 carries a high positive rate while the mid-score cell B_2 carries a low one, which drags global AUC down to 0.234. The thin top cell T still sits above the base, so a gate firing only on T banks $\sigma = 0.2$. The figure illustrates the construction and makes no empirical claim.

The bulk is anti-sorted when $p_1 > p_2$ (equivalently $a + c > 0$): the higher-scored cell B_2 carries the lower positive rate.

Write the AUC as the probability that a random positive outscores a random negative, ties counted as one half, and let $P = b$, $N = 1 - b$ be the total positive and negative mass. Let $(t_1, t_0) = \varepsilon(b + \sigma_0,\ 1 - b - \sigma_0)$ be the positive and negative mass of T, and likewise $(u_1, u_0) = m(p_2, 1-p_2)$ for B_2 and $(v_1, v_0) = m(p_1, 1-p_1)$ for B_1. Summing concordance over ordered cell pairs, where a positive–negative pair is concordant when the positive's cell has the higher score, scores one half within a shared cell, and zero otherwise, gives

$$\mathrm{AUC} \cdot PN = \underbrace{t_1(u_0 + v_0) + u_1 v_0}_{\text{concordant}} + \tfrac{1}{2}\underbrace{\left(t_1 t_0 + u_1 u_0 + v_1 v_0\right)}_{\text{within-cell ties}}. \tag{6}$$

The discordant pairs (a positive scored below its negative, chiefly the large mass $v_1 u_0$ in which a B_1 positive sits below a B_2 negative) contribute zero, and this is where the anti-sorted bulk pays. The two tail terms $t_1(u_0 + v_0)$ are $O(\varepsilon)$. Raising the bulk contrast c, with a following by (5), drives $p_2\downarrow$ and $p_1\uparrow$ and lowers the right-hand side of (6); the contrast is feasible up to $p_2 = b - c \geq 0$. For any fixed ε, σ_0 a large enough bulk deficit therefore dominates the $O(\varepsilon)$ tail surplus, giving $\mathrm{AUC} \leq \frac{1}{2}$ while $\sigma(\tau) = \sigma_0$.

A concrete instance settles existence. Take $b = \frac{1}{2}$, $\sigma_0 = 0.2$, $\varepsilon = 0.1$, and $c = 0.4$, so by (5) $a = 0.356$, giving $p_1 = 0.856$, $p_2 = 0.1$, and a top rate $b + \sigma_0 = 0.7$. Then $t_1 = 0.07$, $t_0 = 0.03$; $u_1 = 0.045$, $u_0 = 0.405$; $v_1 = 0.385$, $v_0 = 0.065$, and (6) evaluates to $0.07(0.405+0.065) + 0.045(0.065) + \frac{1}{2}(0.0021+0.0182+0.0250) = 0.0585$, so $\mathrm{AUC} = 0.0585/0.25 = 0.234 \leq \frac{1}{2}$, while the gate earns $\sigma(\tau) = 0.2$ at firing rate $\phi = 0.1$. □

The construction is not a curiosity (Figure 1). In this situation, a system whose scores are somewhat anti-informative in the bulk (typical of a model which has found some faint, but on-average incorrect, mean-reverting relationship, and so concentrates a few truly valuable signals at the extreme) will report AUC $\leq \frac{1}{2}$ and look worthless, while a conviction gate over the top few per cent is capturing actual edge. A typical scorecard not only underestimates the system but can also give it a failing score. What does register the edge is precision at a fixed firing rate $\pi(\tau)$ at $\phi(\tau)$, the measure reported and used as basis of the attribution in Section 5.5. Of course, this is not an argument for the meaninglessness of AUC. AUC is indeed the right measure for ranking-intensive tasks. It is not the right measure for a selective decision process, and that is why.

3.5 The attribution procedure

The framework will give us a procedure that we apply to the deployed system in Section 5. Given a held-out, walk-forward stream of $(\hat{p}, y)$ pairs and the live ledger of resolved intervals, the decision-value attribution is:

1. Audit deployed uncertainty. On the live ledger, compare nominal interval coverage to empirical coverage. A gap shows that confidence is not self-correcting and fixes the stakes (Section 5.1).
2. Measure the calibration term. Compute ECE before and after recalibration; verify that accuracy and AUC are unchanged, isolating the calibration gap of (3) as the only thing that moved (Section 5.2).
3. Bound the ranking term. Estimate AUC and accuracy directly to establish whether any global ordering skill exists (Section 5.3).
4. Measure the selection term. Tune the conviction threshold $\tau^\star$ on training folds, then report $\pi(\tau^\star)$, $\phi(\tau^\star)$ and $\sigma(\tau^\star)$ out-of-sample with a confidence interval and a test against the base (Section 5.4).
5. Settle the attribution. Express the realised fired precision as $b + \sigma(\tau^\star)$ and attribute each component, contrasting it with the ranking-based reading the standard scorecard would give (Section 5.5).

The output is a single account that tells, in measured numbers, which part of a machine-learning pipeline is doing the paying.

4 The deployed system and the evaluation protocol

The framework is general; the numbers in Section 5 come from one concrete system, which we describe here in enough detail to make the measurements reproducible in spirit. The platform is a live equity decision-support product for the Indian NSE. It runs as two surfaces over a shared model and service layer: a research-facing analytics application and a user-facing web application with a separate forecasting backend. Both read the same trained artefacts, so the numbers in this paper describe the models that real users see.

4.1 Signals, features, and the hybrid ensemble

From the end-of-the-day prices, the system constructs daily features for each of the names, returns over several look-backs, momentum and trend descriptors, mean-reversion indicators, and realised-volatility measures. Then on top of this, it maintains a news layer (as mentioned in Section 4.2) where it scores recent headlines for each entity and utilizes them. These features feed a hybrid ensemble that blends two distinct model families with complementary inductive biases, that are calculated, wherein the first is a gradient-boosted tree ensemble that is designed to capture threshold and interaction structure in the cross-section, and secondly, after that a recurrent sequence model over the price history that is used to capture temporal dependence. Then using a learned blend, both of them are combined, which gives the ensemble releases, per name and horizon, a point forecast, a price interval at a nominal confidence level, and a directional probability $\hat{p}$. Let's say we treat the ensemble as a black box, just for the purpose of this paper. That should not be conisdered as laziness, it is the whole point. It's mainly because the decision-value account does not care how the probability was produced, but only whether it is calibrated and whether the system knows when to trust it. The components that determine those two properties, the calibration stage and the conviction gate, sit downstream of the ensemble, and that is where the analysis concentrates.

4.2 News as a feature: a language model as measurement instrument

The news layer scores each recent headline for an entity and folds the result into the feature stack. Scoring is done by a compact instruction-tuned language model (in the current deployment, Claude Haiku 4.5 from Anthropic), prompted as a quantitative analyst and constrained to emit structured numeric judgements with fixed numerical anchors so that two runs on the same item agree closely. Each scored item is memoised in a content-addressed cache keyed by a hash of the entity and the normalised headline, so a given headline is scored once and reused; when the model is unavailable a transparent lexicon fallback supplies a conservative score that is flagged for downstream consumers. We describe the news layer because it is part of the deployed system and because the journal asks that any use of AI tools be disclosed (see also the declaration at the end of the paper). We do not, however, claim a standalone return for it: the live ledger does not store per-event return labels, so an isolated news alpha is not identifiable from our deployment, and we decline to reconstruct one. Its role in this paper is as one input to the ensemble whose end-to-end behaviour we measure.

4.3 Calibration stage

A directional probability is only useful if it is calibrated: among names assigned $\hat{p} \approx 0.7$, about 70% should rise. We measure calibration by the expected calibration error over B equal-width bins,

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{|G_b|}{N} \left| \overline{y}_b - \overline{p}_b \right|, \tag{7}$$

where $\overline{p}_b$ and $\overline{y}_b$ are the mean predicted probability and mean realised outcome in bin b, complemented by the Brier score $\frac{1}{N}\sum_i (\hat{p}_i - y_i)^2$ [Brier 1950]. The calibration stage applies a learned monotone map to the ensemble's raw directional score. Section 5.2 documents what happened when we first measured (7) on the deployed model. It was badly broken, and the bug was the system's own.

4.4 The conviction gate

Calibration makes $\hat{p}$ honest; it does not make it skilful. If short-horizon single-name direction is near-random, a calibrated probability sits near the base rate almost everywhere, and acting on every name is pointless. The conviction gate is the one-sided selective rule d_τ of Section 3.1: for horizon H it emits an UP call when $\hat{p} \geq \tau^\star$ and NEUTRAL otherwise. The threshold is tuned on training folds as the smallest τ whose fired bucket reaches a target precision p_0 while still firing at least a minimum fraction ϕ_0 of the time,

$$\tau^\star = \min\{\tau : \ \pi(\tau) \geq p_0 \text{ and } \phi(\tau) \geq \phi_0\}, \tag{8}$$

with $p_0 = 0.60$ and $\phi_0 = 0.05$ in deployment. The gate is intentionally one-sided for the empirical reason given in Section 5.3. Choosing $\tau^\star$ on training data and reporting π, ϕ out-of-sample is what keeps the selection-value estimate honest; this is the threshold-tuning analogue of not peeking at the test set.

4.5 The append-only audit ledger

Since April 2026 the deployed system has written every issued forecast to an append-only ledger: entity, issue timestamp, target date, point forecast, interval, nominal confidence, directional probability and horizon. Each row is later stamped with the realised price and outcome on its original target date and is never retro-edited; scoring is first-look only. The ledger matters for two reasons. It is the strongest evidence we have, because it is produced by the running system rather than reconstructed after the fact, and it is the mechanism that makes the system falsifiable in deployment: a forecaster that cannot be confronted with the future it actually faced cannot be audited, and an un-auditable forecaster in a money-bearing setting is the thing the backtest-overfitting literature warns about [Bailey et al. 2017].

4.6 Data and evaluation protocol

Two bodies of evidence are kept strictly separate. The first is an offline walk-forward study on daily data from 2018 to 2026 over a 54-name large-cap universe, used to fit and evaluate the ensemble, the calibration stage and the gate. All offline statistics are out-of-sample under an expanding window, so no future information enters a fitted fold; the gate threshold is tuned only on training folds. The second is the live append-only ledger of Section 4.5. We report, from the offline study, the calibration repair, the accuracy ceiling and the selective signal; from the live ledger, interval coverage. No quantity in Section 5 is illustrative unless its caption says so. Figure 2 places the components in one pipeline and marks where each measurement is taken.

Two further points bear on how far the numbers can be trusted. Both deployment surfaces of Section 4 read the same trained artefacts, so the figures below describe what real users see rather than a research-only variant. And the offline study is reproducible in the sense an audit cares about. The universe is fixed in advance; the walk-forward windows expand rather than slide, so no fitted fold sees its own future; the gate threshold $\tau^\star$ is selected on training folds alone; and every persisted statistic traces to a stored artefact rather than to a number typed into the manuscript. The live ledger adds the stronger guarantee that the running system, not a reconstruction, produced each forecast, and that the score against the outcome is taken on the original target date and never revised afterwards.

5 Results: an account of where the value comes from

We now run the attribution procedure of Section 3.5 on the deployed system, in order.

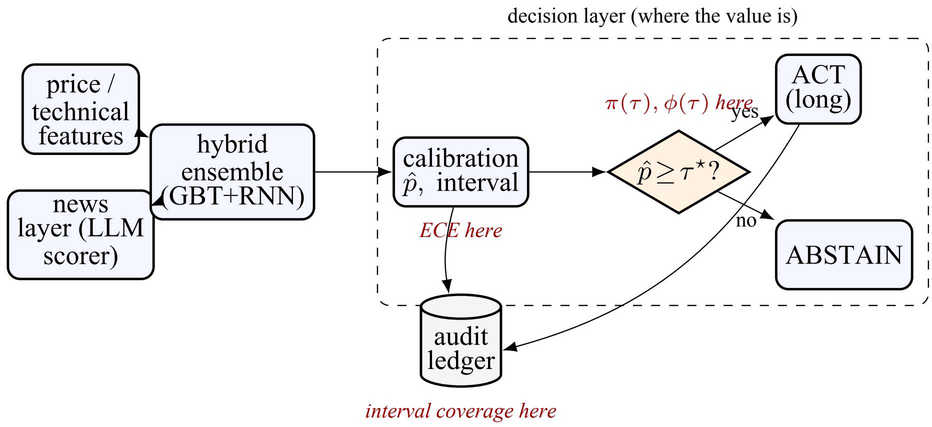

Figure 2: The deployed pipeline and where each measurement in Section 5 is taken. Features (price and news) feed a hybrid ensemble; its directional probability is calibrated, then passed to a conviction gate that acts or abstains; every issued forecast is written to an append-only ledger and later scored against its target date. The dashed box is the decision layer the attribution finds responsible for the realised value. The schematic is conceptual.

Table 1: Live-ledger interval coverage by horizon (NSE large caps, 19 April–12 June 2026). All intervals carry a nominal 90% level.

Horizon	Resolved forecasts	Nominal coverage	Empirical coverage
5 trading days	8	90.0%	100.0%
10 trading days	575	90.0%	69.2%
20 trading days	38	90.0%	86.8%
All	621	90.0%	70.7%

5.1 Deployed uncertainty does not self-correct

We start with the live ledger because it sets the stakes. Between 19 April and 12 June 2026 the system issued and later resolved 621 price-interval forecasts on seven NSE large caps (ABB, HDFCBANK, ICICIBANK, INFY, RELIANCE, SBIN, TCS), each carrying a nominal 90% interval with a median half-width of 5.6% of price. Were the intervals well calibrated, about 90% should contain the realised price. They did not: empirical coverage was 70.7% overall, a 19.3 percentage-point shortfall. Most forecasts were at the ten-day horizon, where coverage was 69.2% over 575 resolved forecasts; the smaller twenty-day and five-day buckets covered 86.8% and 100% (Table 1, Figure 3). Put plainly, realised prices broke through the $\pm 5.6\%$ band far more often than one time in ten, so the deployed model's stated confidence was systematically overstated, and nothing in the pipeline noticed on its own. This is the empirical reason calibration and abstention cannot be afterthoughts, and it is step one of the attribution: before any value can be claimed, the uncertainty has to be shown to be untrustworthy as shipped.

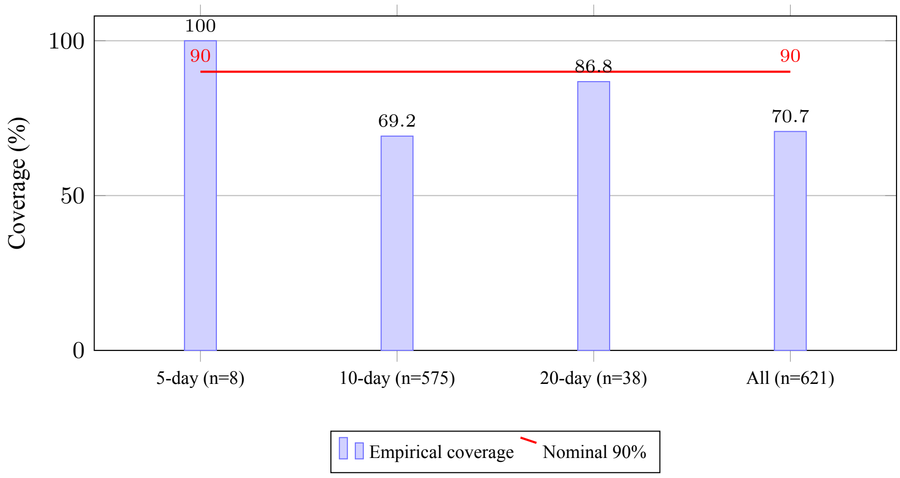

Figure 3: Empirical coverage of nominal 90% price intervals on the live ledger, by horizon. The deployed intervals are materially under-covered at the dominant ten-day horizon, which carries 575 of the 621 forecasts.

5.2 The calibration term: a stacking bug, and a three-stage repair

The offline diagnosis of overconfidence is sharper still; this and the two results that follow are walk-forward measurements, not live-ledger figures. The deployed model's next-day directional head reported probabilities clustered near 0.80 against a realised accuracy of about 0.51, giving ECE $= 0.345$ and a Brier score above 0.366, worse than always predicting 0.5. Tracing the pipeline exposed a textbook stacked-generalisation trap [Wolpert 1992]: an isotonic calibrator had been fitted on the out-of-fold average of the base learners but then applied to the stacked prediction, a different distribution, and the final blend carried no calibration of its own. The fix is mechanical. We refit the isotonic map on the same base-average it was meant to correct, form the blend, then fit a final Platt scaler [Platt 1999] on a held-out fold and apply it to both the test rows and the live row. This cut ECE from 0.345 to 0.049 and pulled the reported up-probability back to a realistic neighbourhood of 0.50.

The crucial observation, and the reason this is step two of the attribution rather than a contribution in its own right, is that accuracy did not move. The recalibration map is monotone, so by the argument of Section 3.2 it cannot change the model's ordering of names, its accuracy, its AUC or its conviction lift. It changed only the calibration gap, pulling the advertised probability onto the delivered one. In decision-value terms, calibration did not create value; it made the value that selection would later extract collectible, by ensuring that the threshold $\tau^\star$ would mean what it was set to mean. A system shipped at ECE $= 0.345$ would have a conviction gate firing on a probability scale that lies by 30 points; the gate would be tuned on a fiction. The episode is also a clean, transferable lesson: a calibrator is bound to the distribution it was fitted on, and stacking quietly creates more than one such distribution.

5.3 The ranking term is essentially zero

Once calibration was repaired, it exposed an uncomfortable fact that the inflated probabilities had hidden. With $\hat{p}$ pulled back to the base rate, we asked directly whether any robust single-name directional edge survives at short horizons. A cross-sectional model predicting five-day direction across the universe reached 50.09% out-of-fold accuracy on 131,809 ticker-days, with ranking AUC near 0.49, which is chance. We confirmed the result five ways, varying the learner and the feature set and gating by trend and by market regime: a per-ticker ensemble, a cross-sectional logistic model, a gradient-boosted model with reversion features, trend/momentum gating, and market-regime gating. None produced a stable edge over the always-up base rate at the daily scale; in this sample the trend-gated variants sat below the base rate, the universe being mildly mean-reverting at that horizon. This is consistent with the efficient-market prediction for liquid names [Fama 1970, Malkiel 2003], and it is step three of the attribution: the ranking term that accuracy and AUC are built to reward is, here, absent. By Proposition 1, that absence does not by itself condemn the system (an edge can still live in the tail), but it does mean that any value we find must be a tail property, and cannot be a ranking one.

5.4 The selection term carries a small, stable edge

Lengthening the horizon to thirty trading days (about a month and a half, the use case the product is actually sold for) opens two modest edges: the upward drift of equities, since about 58% of thirty-day windows are up, and a small idiosyncratic component that a calibrated cross-sectional model can find on momentum, trend, reversal and volatility features. Walk-forward from 2018 to 2026 over the 54-name universe (91,471 training rows, 50,272 out-of-sample rows), the conviction gate of (8) settles at $\tau^\star = 0.63$. Its fired UP bucket achieves 60.6% directional precision against a 58.0% always-up base,[1] a selection value of $\sigma(\tau^\star) = +2.6$ percentage points, while firing on only 5.65% of observations; the gate abstains on the remaining 94%. The fired bucket pools to 2,840 calls, so the 60.6% precision carries a 95% binomial interval of 58.8–62.4%; a one-sided test against the 58.0% base gives $z \approx 2.8$ ($p \approx 0.003$), so the pooled edge is distinguishable from the base at conventional levels. Whether this clears cost is the net-of-cost question of (2). We lack position-level fills to compute V exactly, but the order of magnitude is reassuring: at a thirty-day horizon the typical absolute move in these names runs to several per cent, against a round-trip cost of order 0.2% for liquid large caps, so a long/abstain edge of 2.6 points is not erased by friction. A precise V awaits position-level accounting, which we flag among the limitations.

We aren't reading too much into it. The fired numbers per year are low (Table 2), so the year-to-year swings, from $+10.8$ points in 2025 to -2.4 in 2024, are within sampling noise, and a fair summary is a small but repeatable tail edge rather than a strong one. The gate beats its base rate in 4 out of 5 test years (Figure 4). The only exception, 2024, sits in the table with the rest. The ranking AUC of this signal is about 0.47, at or below chance, which is the regime Proposition 1 describes: the edge lives in the high-conviction tail and not in the ordering, so precision at a fixed firing rate registers it while AUC overlooks it. Confident DOWN calls, for completeness, reach only about 10% precision

[1] The pooled base of 58.0% is the unconditional out-of-sample up-rate over all $50{,}272$ rows (the always-act comparator b of Definition 1), not the fires-weighted mean of the per-year bases in Table 2, which is 55.9%. The unconditional rate is the more conservative comparator; measured against the fires-weighted base the pooled edge would read $+4.7$ points instead of $+2.6$.

Table 2: Walk-forward selective signal, by test year. Precision is the realised up-rate of the fired high-conviction bucket; base is the always-up rate that year; fires is the number of high-conviction calls that year.

Test year	Fires	Fired-bucket precision	Always-up base	Edge (pp)
2022	365	61.6%	54.0%	+7.6
2023	290	68.3%	65.3%	+3.0
2024	932	53.3%	55.7%	−2.4
2025	822	69.0%	58.2%	+10.8
2026	431	54.5%	47.1%	+7.4
Pooled	2,840	60.6%	58.0%	+2.6

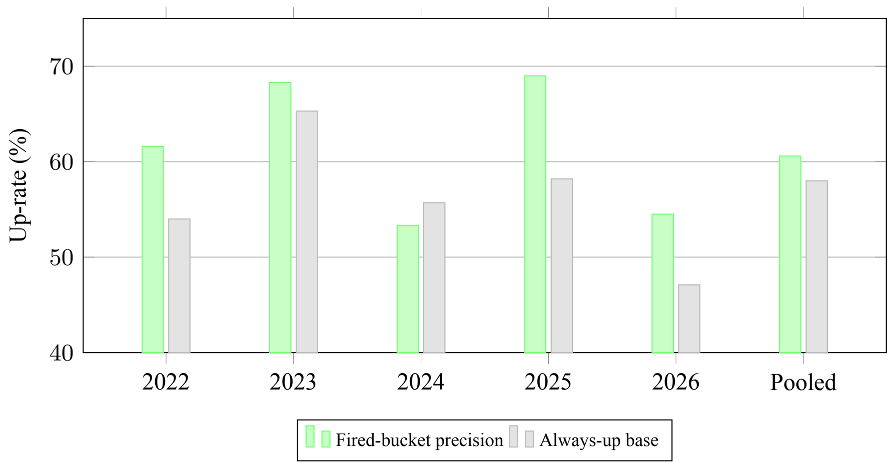

Figure 4: Selective signal versus its base rate by test year. The conviction gate beats the always-up base in four of five years and pooled, while firing on about 5.6% of observations.

out-of-sample, destroyed by drift, which is why the gate is one-sided and never emits them.

5.5 Settling the attribution

We can now state, in measured numbers, where the realised decision value comes from. The system's only positive, collectible edge is the fired precision of its conviction gate, $\pi(\tau^\star) = 60.6\%$. By the decomposition (3) and Definition 1 this splits cleanly into the components the four preceding subsections measured, summarised in Table 3 and drawn as a waterfall in Figure 5.

As it is visible, the split is really clean. 58.0 of the 60.6 points, of realised fired precision, are a drift (a market beta any index fund collects), and the rest of the 2.6 points, are the selection value that the system adds. The ranking term that accuracy (50.0%) and AUC (≈ 0.47) measure contributes nothing, maybe on the standard scorecard this

Table 3: Decision-value attribution of the deployed system. Each row is a measured quantity from Sections 5.1–5.4; the realised fired precision is the sum of the drift base and the selection value, with ranking and calibration acting as described.

Component	Measured value	Role in decision value
Drift (base rate b)	58.0% up-rate	Beta, not skill; available with no model.
Selection value $\sigma(\tau^\star)$	$+2.6$ pp	The only positive, collectible edge; a tail property.
Ranking skill (AUC$-\frac{1}{2}$)	-0.01 to -0.03	Daily ≈ -0.01, 30-day ≈ -0.03; invisible to selection (Prop. 1).
Calibration (ECE)	$0.345 \to 0.049$	Enabling, not additive; makes $\tau^\star$ trustworthy.
Deployed interval coverage	70.7% vs 90% nominal	The risk the attribution exists to expose.
Realised fired precision $\pi(\tau^\star)$	$60.6\% = b + \sigma$	What the system can actually be paid for.

signal is seen as a failure, and a small but statistically distinguishable edge would be discarded by a practitioner who graded it that way. The term calibration does not appear as an additive term anywhere to be fair, just as Section 3.2 in this paper predicts, and yet without the repair from ECE $= 0.345$ to 0.049 the conviction threshold would have been built on a lying probability scale and the 2.6-point edge would not have been collectible as such. Hence, the account says three things at once, that is, the model's prediction is not where the value sits, but selection is in it's place; and calibration is the precondition that lets the selection edge be banked. Table 3 captures the economics of this system in values, and the pattern, we argue next, is not unique to it.

6 Discussion

6.1 What the attribution generalises to

The numbers are from one platform; but the shape of the attribution is not. In a domain, any deployed forecaster that decides whether to act, where the base rate is high and the ordering signal is weak, will show a similar profile, which is basically a large beta-like base, then a thin selection edge in the tail, and a calibration term that enables rather than adds, and lasltly a ranking term which the standard scorecard over-weights. Markets are the cleanest example here, because their efficiency drives the ranking term towards zero [Fama 1970, Malkiel 2003] and their outcomes are public and unforgiving, but the structure recurs, wherever while evaluating, a model evaluates an incoming stream of opportunities and is judged only on the few specific picks it selects. The recommendation that follows is concrete and portable, and we set it out as a reporting checklist in Section 6.4. A great deal of financial-ML work optimises the ranking term, which our results suggest is the one term least likely to survive deployment.

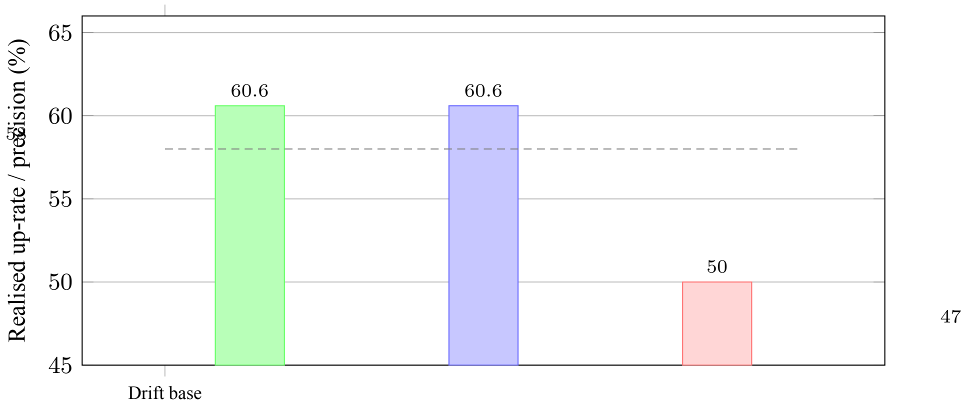

Figure 5: Decision-value waterfall. The drift base (58.0%) plus the selection value of the conviction gate (+2.6 pp) account for the realised fired precision (60.6%). The two right-hand bars show what the conventional scorecard reports for the same signal, and are not up-rates: directional accuracy at chance (50.0%) and a ranking AUC of about 0.47 (plotted as AUC×100 for comparison), neither of which sees the tail edge. The dashed line marks the drift base. All values are measured (Sections 5.3–5.4).

6.2 Why the standard scorecard misleads, restated

Proposition 1 carries the generalisation. Accuracy and AUC summarise global behaviour. A selective system is defined by what happens in a thin tail. The two are provably decoupled. This is not a quirk of our data. It is why a literature that reports AUC can simultaneously be full of models that look skilful and empty of models that deploy: the metric rewards a property (global ordering) that is both easy to overfit in backtests [Bailey et al. 2017, Harvey et al. 2016] and largely irrelevant to a system that acts on its tail. Reporting precision at coverage, as the selective-prediction community does [Geifman and El-Yaniv 2017, El-Yaniv and Wiener 2010], aligns the metric with the deployment. The decision-value account grades a model on the trades it would actually make.

6.3 Threats to validity and limits

Emphasis is deserved to the five limits. Very firstly, the live ledger is young, to be precise just about two to three months and only about seven names, so its, mathematically, a 19.3-point coverage shortfall, though at the dominant horizon, it is large and consistent, and is a current snapshot rather than a stationary law, and also the directional-probability sample on the ledger is still too small (about 28 resolved next-period calls) to support a, honest live reliability diagram, so our live claim therefore rests on the 621-forecast interval-coverage evidence we mentioned, and the calibration and selection results are purely offline walk-forward measurements. Secondly, the selection edge itself is small in absolute terms (which came to be 2.6 points) and its per-year variation is within sampling noise; we report it explicitly not as a strong signal, but as a thin, repeatable tail edge honestly. Third, the news layer's contribution is argued from its place in the pipeline and not from an isolated backtest, because the deployment does not store per-event return labels; we prefer to say this plainly than to reconstruct a clean number, which is the kind of exercise the backtest-overfitting critique exists to catch. Fourth, the calibration drifts over time and must be maintained on a schedule; we treat the maintenance question as out of scope here and a matter for separate work, and make no claim about an optimal recalibration cadence in this paper. Fifth, we attribute the cost-free selection value σ rather than the full net-of-cost decision value V of (2); the order-of-magnitude argument of Section 5.4 suggests the edge survives friction, but a position-level computation of V from realised fills and sizing is left for future work.

6.4 Reporting decision value: a short checklist

The attribution is easy to adopt, so we state it as a checklist a practitioner can run against any deployed selective forecaster. Firstly, issue a coverage audit. Take live ledger, nominal and empirical interval coverage comparison. Report gap by horizon (as done in Table 1). Report calibration pre and post any post-hoc fixing and post post-hoc fixing. Report expected calibration error with the remark that the accuracy did not change; this confirms fix only altered reliability. Then provide explicit statement of the base rate b and compare the fired bucket to it (not to an equal odds prior); equity long-only has large drift; that is not skill. Report precision at a specified firing rate $\pi(\tau)$ at $\phi(\tau)$, report a confidence interval for it and a test against b instead of accuracy or AUC. Break out the ranking term: provide the AUC, and if it is around or below 0.5 and tail precision is greater than 0, state this; that is the footprint of a tail-only edge. Finally, where costs are known, convert the selection value to a net-of-cost figure through (2); where they are not, scope the claim to the cost-free core and name the quantity being reported. A paper that fills in these six lines is hard to misread, and a system that cannot fill them in is not yet ready to be trusted with a position.

6.5 Relation to selective and conformal prediction

The conviction gate is an instance of selective classification [Chow 1970, Geifman and El-Yaniv 2017]; what the decision-value account adds is the attribution, separating the gate's selection value from the drift it rides on and the calibration that makes it bankable, together with the tail–AUC result that justifies measuring it at a fixed firing rate. The interval under-coverage we documented in Section 5.1 is the failure that distribution-free methods are built to address; adaptive conformal inference [Gibbs and Candès 2021, Angelopoulos and Bates 2021] is designed for this kind of non-stationary coverage shortfall, and replacing the deployed parametric intervals with an adaptive conformal layer is the clearest next step the attribution points to.

7 Conclusion

Machine-learning systems for markets are usually graded on accuracy and AUC, and most of them, graded carefully, have neither. We have argued that this is the wrong way to keep score. A deployed forecaster is paid for the quality of the positions it actually takes, weighed against those it skips, and we made that quantity, its decision value, precise by extending the calibration–refinement and reliability–resolution decompositions of forecast verification into the abstaining, cost-aware decision setting. The framework comes with a theorem that explains why the usual scorecard cannot see what such a system is worth: selection value lives in the tail of the score distribution and is provably decoupled from global AUC, so a system with real decision value can be reported at chance. We then ran the attribution on a live equity platform. The result is uncomfortable, and we believe it is representative. Deployed 90% intervals covered only 70.7% of outcomes, so confidence had to be earned rather than assumed; a calibration repair cut expected calibration error from 0.35 to 0.05 without adding a point of accuracy; single-name direction proved unpredictable beyond drift; and a conviction gate that abstains on 94% of names nonetheless banked a 60.6% hit rate against a 58.0% base, in four of five years. Drift is just beta. The edge comes from selection. Calibration is the precondition that makes it bankable. And prediction, the thing the field optimises for, contributes almost nothing. The practical message follows: instrument the uncertainty layer, grade the model on the trades it would make, and put the engineering effort where the attribution says the value is. Replacing the deployed intervals with an adaptive conformal layer, and a per-event labelled evaluation of the news contribution, are the clear next steps.

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Declaration of generative AI and AI-assisted technologies

Two distinct uses are disclosed. (i) As an instrument within the system under study: an instruction-tuned large language model (Claude Haiku 4.5, Anthropic) scores news headlines for the feature layer described in Section 4.2, from the anchored prompt summarised there; these outputs are part of the deployed pipeline, were validated against a deterministic lexicon fallback, and are persisted for audit. (ii) In manuscript preparation: during the writing of this paper the authors used a large language model solely to improve the language and readability of author-written drafts. No generative tool produced data, results, figures or interpretations; every quantitative finding derives from the authors' deployed system and its measured ledger. After using these tools the authors reviewed and edited the content and take full responsibility for the publication.

Data availability

The live forecast ledger and the offline walk-forward statistics underlying Tables 1–3 and Figures 3–5 can be made available by the corresponding author on reasonable request, subject to the platform's terms. Source price data are public NSE end-of-day records.

Conflict of interest

The authors declare that they have no conflict of interest.

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